



Question(s): 6/9

Geneva, 06-14 Sep 2022

CONTRIBUTION**Source:** Ministry of Communications, India**Title:** Proposed A.1 justification for proposed draft new ITU-T Recommendation J.BD2M "Functional requirements for terminal devices of the Broadcast Direct to Mobile (D2M) networks"**Contact:** Avinash Agarwal
Ministry of Communications
IndiaTel: +91-9868171222
E-mail: avinash.70@gov.in**Contact:** Meenakshi Singhvi, Prasar Bharati
Ministry of Communications
IndiaTel: +91-9426302816
E-mail: meenakshi.singhvi@prasarbharati@gov.in**Abstract:** This contribution contains the proposed A.1 justification, placed as Annex-1, for proposed draft new ITU-T Recommendation on "Functional requirements for terminal devices of the Broadcast Direct to Mobile (D2M) networks".

The proposed Recommendation will specify the functional requirements of smartphones and similar terminal devices that can receive live (linear) Digital Terrestrial Transmission (DTT) signals and that may support leading Direct to Mobile (D2M) TV transmission standards.

While mobile handsets are now the most preferred terminal devices for viewing audiovisual content, they generally are not capable of receiving the broadcast signals directly and users have to rely on content streamed as data using the mobile network (LTE, 5G).

The aim of this proposed Recommendation is to promote design and manufacturing of such smartphones and similar terminal devices that can directly receive broadcast signals.

Annex 1

A.1 justification for proposed draft new ITU-T J.BD2M "Functional requirements for terminal devices of the Broadcast Direct to Mobile (D2M) networks"

Question:	Q6/9	Proposed new ITU-T Recommendation	Online or Geneva, 06-14 September 2022	
Reference and title:	ITU-T J.BD2M "Functional requirements for terminal devices of the Broadcast Direct to Mobile (D2M) networks"			
Base text:	SG9-Cnnn		Timing:	2023-08
Editor(s):	Avinash Agarwal, Ministry of Communications, India avinash.70@gov.in Meenakshi Singhvi, Prasar Bharati, Ministry of Communications, India meenakshi.singhvi@prasarbharati@gov.in		Approval process:	AAP
Scope (defines the intent or object of the Recommendation and the aspects covered, thereby indicating the limits of its applicability): The proposed Recommendation will specify the functional requirements of smartphones and similar terminal devices that can receive live (linear) Digital Terrestrial Transmission (DTT) signals and that may support leading Direct to Mobile (D2M) TV transmission standards. While mobile handsets are now the most preferred terminal devices for viewing audiovisual content, they generally are not capable of receiving the broadcast signals directly and users have to rely on content streamed as data using the mobile network (LTE, 5G). The aim of this proposed Recommendation is to promote design and manufacturing of such smartphones and similar terminal devices that can directly receive broadcast signals.				
Summary (provides a brief overview of the purpose and contents of the Recommendation, thus permitting readers to judge its usefulness for their work): With the advent of new broadcasting standards especially in digital domain, the challenges and responsibility go hand in hand to pay focused attention in developing compatible, inexpensive, easy to use, comfortable methods of viewing that content on receiver sets. The user experience (UX), and the role of a well-designed user interface (UI) are important for adoption and user engagement. In the past, also various technologies fell short of this desired goal due to the lack of compliant receiving systems at the end user. All the improvements and upgrading in the transmission/ broadcast technology should also address the receiving system mechanism in parallel. While mobile handsets are now the most preferred terminal devices for viewing audiovisual content, they generally are not capable of receiving the broadcast signals directly and users have to rely on content streamed as data using the mobile network (LTE, 5G). This Recommendation aims to specify the functional requirements of smartphones and other similar terminal devices that can receive live (linear) Digital Terrestrial Transmission (DTT) signals and that may support leading Direct to Mobile (D2M) TV transmission standards. The aim is to promote design and manufacturing of such smartphones and devices that can directly receive broadcast signals. A brief overview of the features proposed to be covered in this Recommendation is as follows: The terminal device should be designed to deliver high performance in static, mobile and multipath-prone environments. The requirements should ensure availability of reliable reception capability in the mobile service environment, especially where more and more consumers use their mobile handsets as a primary video services device. Small form factors of smartphones and other portable devices pose challenges such as small embedded antenna requirements, limited battery operation, receiver integration in the form of a chip, different user interfaces, and more.				

These smart phones may have an embedded antenna or an external one. The D2M receiving system may be embedded into the mobile handset in the form of a chip or an external device (dongle) can be utilised. The dongle may be a low power USB connected device supporting different operating systems of the handsets. A small, low power consumption, software defined radio chip may be most suitable for initial deployments. Reference devices of this variety have already been built and demonstrate the ability to support a D2M service. When there is a true ‘mass market’ (multiple millions of devices) the D2M functionality may become an embedded element of a mobile device Services on a Chip (SoC). One of the requirements of having a TV band chip (D2M) in the mobile device is to ensure that it can receive audiovisual content through both broadcast (D2M) and telecom (4G, 5G) signals. This may require interconnect and interoperability within the network environment. To support D2M signals, these devices with embedded chips should work under frequency ranges allocated for Digital Terrestrial Transmission (DTT) in the UHF Band with input channel bandwidths of 5/6/7/8 MHz. This will necessitate relevant band filtering to preclude adjacent services from interfering (i.e. LTE). Beyond the physical requirements that provide for reliable reception, there will likely exist mandatory provisions to support local emergency alerts, regional disaster management updates and nationwide broadcasts of strategic and national importance to all compatible smart devices/ smartphones.
Relations to ITU-T Recommendations or to other standards (approved or under development): None
Liaisons with other study groups or with other standards bodies: None
Supporting members that are committing to contributing actively to the work item: Ministry of Communications, India